



Futures Markets

Futures Contracts



Available on a wide range of assets



Exchange traded



Specifications need to be defined:

What can be delivered,
Where it can be delivered, &
When it can be delivered

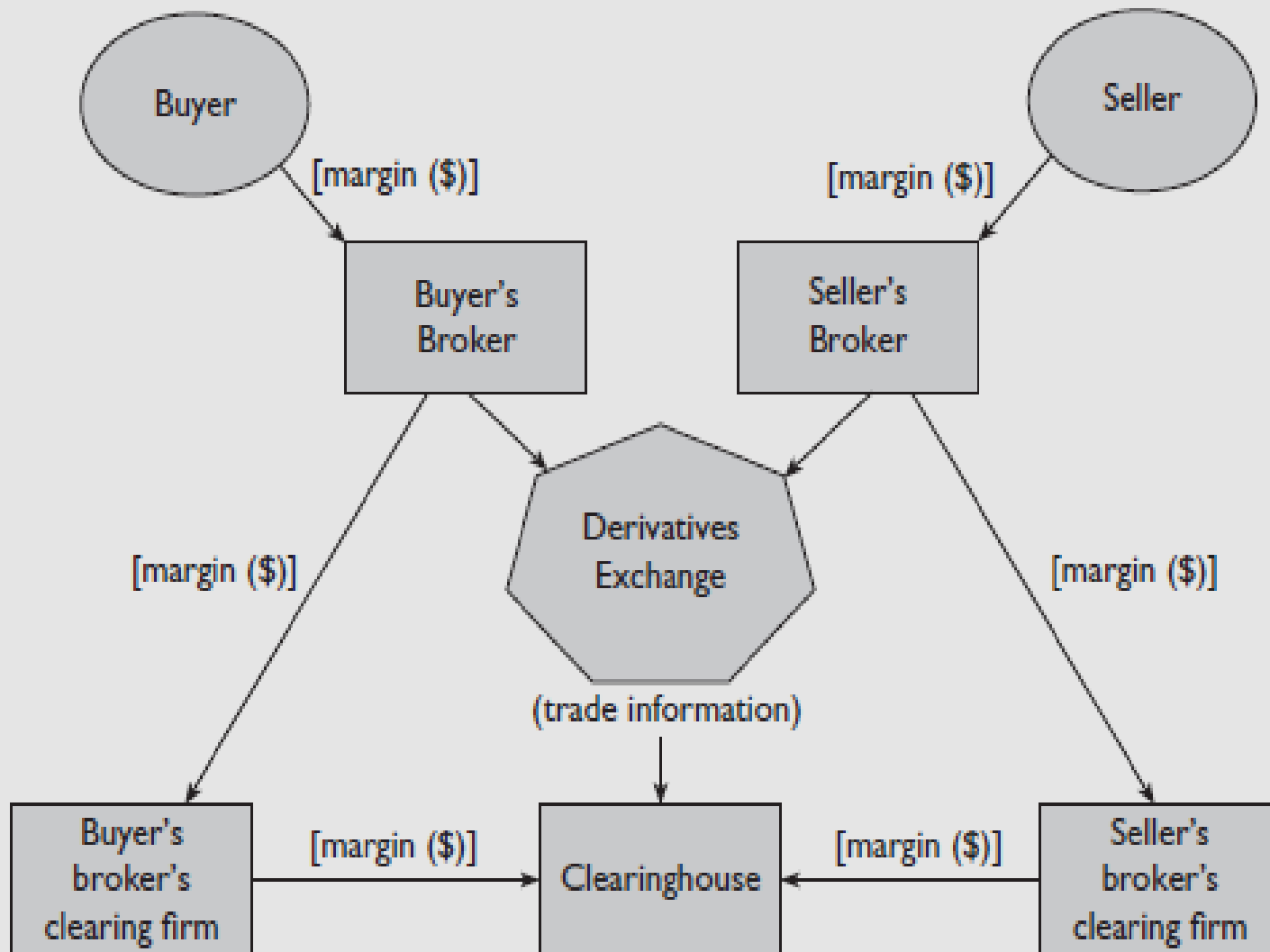


Settled daily

Futures Products in India

- Equity Index Futures
- Single Stock Futures
- Interest Rate Futures
- Commodity Futures
- Currency Futures
- Volatility Futures

Future Trading



Future Price Quotation

INSTRUMENT TYPE	EXPIRY DATE	OPTION TYPE	STRIKE PRICE	OPEN PRICE	HIGH PRICE	LOW PRICE	CLOSE PRICE	PREV. CLOSE	LAST PRICE	CHNG	%CHNG	VOLUME (Contracts)	VALUE (₹ Lakhs)
 Stock Futures	24-Sep-2020	-	-	2,310.60	2,339.50	2,290.75	2,320.10	2,316.30	2,317.25	0.95	0.04	65,776	7,70,757.84

Market Depth - Order Book

QTY	BID (₹)	ASK (₹)	QTY	TRADE INFORMATION		OTHER INFORMATION	
4,040	2,317.25	2,318.00	505	Traded Volume (Contracts)	65,776	Daily Volatility	2.70
4,040	2,317.20	2,318.40	505	Traded Value (₹ Lakhs)	7,70,757.84	Annualised Volatility	51.52
505	2,317.05	2,318.50	505	VWAP	2,320.38	Implied Volatility	-
2,020	2,317.00	2,319.00	505	Market Lot	505	Settlement Price	2,320.10
505	2,316.80	2,319.15	505	Open Interest (Contracts)	64,218	Client Wise Position Limits	-
Buy (Total Quantity) -		Sell (Total Quantity) -		Change in Open Interest	-3,005	Market Wide Position Limits	65,49,49,809
8,45,875		12,86,740		% Change in Open Interest	-4.47		

Intraday Future Price



Margins and M2M

- A margin is cash or marketable securities deposited by an investor with his or her broker
- The balance in the margin account is adjusted to reflect daily settlement
- Margins minimize the possibility of a loss through a default on a contract

Margin Cash Flows

- A trader has to bring the balance in the margin account up to the initial margin when it falls below the maintenance margin level
- A member of the exchange clearing house only has an initial margin and is required to bring the balance in its account up to that level every day.
- These daily margin cash flows are referred to as variation margin
- A member is also required to contribute to a default fund

Example of a Futures Trade

- An investor takes a long position in 2 December gold futures contracts on June 5
 - contract size is 100 oz.
 - futures price is US\$1,250
 - initial margin requirement is US\$6,000/contract (US\$12,000 in total)
 - maintenance margin is US\$4,500/contract (US\$9,000 in total)

A Possible Outcome

Day	Trade Price (\$)	Settle Price (\$)	Daily Gain (\$)	Cumul. Gain (\$)	Margin Balance (\$)	Margin Call (\$)
1	1,250.00				12,000	
1		1,241.00	-1,800	- 1,800	10,200	
2		1,238.30	-540	-2,340	9,660	
.....						
6		1,236.20	-780	-2,760	9,240	
7		1,229.90	-1,260	-4,020	7,980	4,020
8		1,230.80	180	-3,840	12,180	
.....						
16	1,226.90		780	-4,620	15,180	

ACC SEP FUT :

16/09/2020 09:35

1388

VOLUME

14.0K

OPEN: 1388.9

CLOSE: 1388

HIGH: 1389.05

LOW: 1386.25

ZERODHA

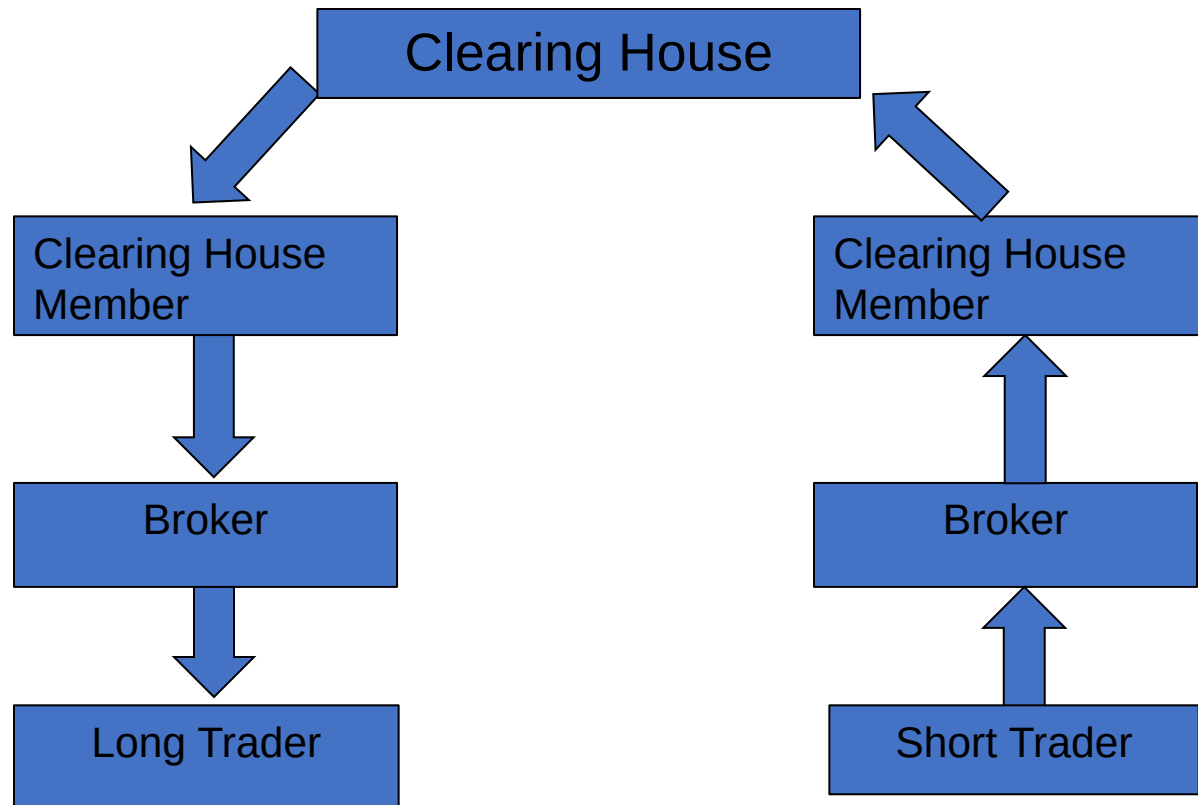
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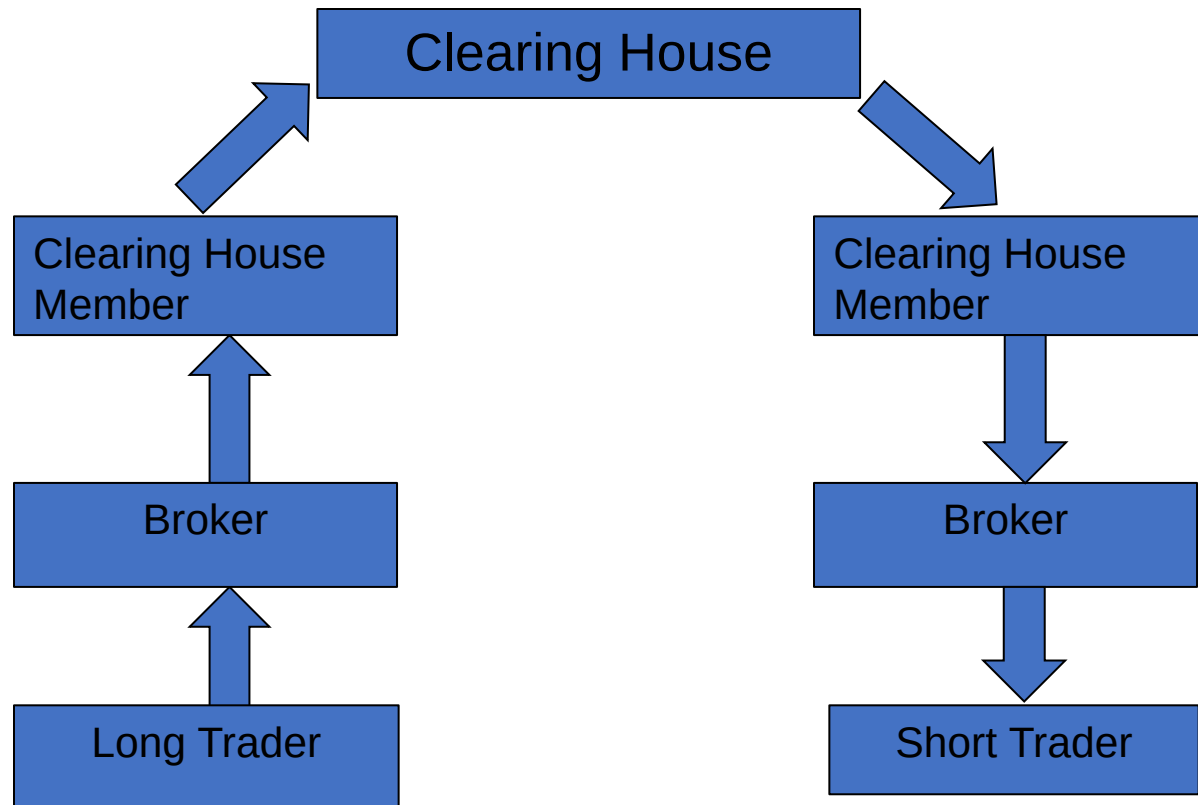
1392.75

78.07

Margin Cash Flows When Futures Price Increases



Margin Cash Flows When Futures Price Decreases



Intriguing Questions???

1. How to decide the initial margin requirement for a contract?
2. What are the factors influence the decision of initial margin level?
3. What are the components of initial margin?

Types of Orders

- **Market** - A market order is an order to buy or sell scrips at the current best available price.
- **Limit:** A limit order allows you to buy or sell a stock at the price you have set or a better price.
 - In other words, if you place a buy limit order at Rs 92, you want to buy the stock from the exchange only at Rs 92 or lower. You don't want to pay more than Rs 92. Similarly, if you place a sell limit order at Rs 95, you want to sell the stock at Rs 95 or higher.
 - The advantage of placing a limit order is that you can place buy/sell order at the desired price. However, there is a chance that your order may not get filled partially or completely depending upon if a counter order is available for some quantity or none at the price you've specified.
- **Stop-loss order:** A stop-loss order is a buy/sell order placed to limit the losses when you fear that the prices may move against your trade. For instance, if you have bought a stock at Rs 100 and you want to limit the loss at 95, you can place an order in the system to sell the stock as soon as the stock comes to 95. Such an order is called 'Stop Loss', as you are placing it to stop a loss more than what you are ready to risk.

➤ **Stop-Loss Limit Order (SL):**

- Case 1 > if you have a buy position, then you will keep a sell SL
- Case 2 > if you have a sell position, then you will keep a buy SL

In Case 1, if you have a buy position at 100 and you wish to place an SL at 95.

You will place a Sell SL order with price and trigger price. Since your order needs to be triggered first, the (trigger price $\geq$ price.) Here, this order type gives you a range of the Stop-Loss.

Let's assume a range of Rs 0.10 (10 paise). Here, you can keep trigger price = 95 and price = 94.90.

When the price of 95 is triggered, the sell limit order is sent to the exchange and your order will be squared off at the next available bid above 94.90. So, your SL order may get executed at 96 or 94.95 but not below 94.90.

- You will place a Buy SL order with price and trigger price. Since your order needs to be triggered first, (the trigger price $\leq$ price.) Here, this order type gives you a range of the stop-loss.
- Let's assume a range of Rs.0.10 (10 paise). Here, you can keep trigger price = 105 and price = 105.10. When the price of 105 is triggered, the buy limit order is sent to the exchange and your order will be squared off at the next available offer below 105.10. So, your SL order may get executed at 105.05 or 105 but not above 105.10.

Forward Contracts vs Futures Contracts

FORWARDS	FUTURES
Private contract between 2 parties	Exchange traded
Non-standard contract	Standard contract
Usually 1 specified delivery date	Range of delivery dates
Settled at end of contract	Settled daily
Delivery or final cash settlement usually occurs	Contract usually closed out prior to maturity
Some credit risk	Virtually no credit risk

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